

ALEXANDER RYABCHENKO

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PROFILE

PhD student at the University of Toronto and Vector Institute researching the statistical and computational foundations of modern machine learning, including reinforcement learning, optimization, and post-training.

EDUCATION

Doctor of Philosophy (PhD), Statistical Sciences <i>University of Toronto</i>	2026 – Present Toronto, ON
• Advisors: Murat A. Erdogdu, Wenlong Mou, Stanislav Volgushev.	
Honours Bachelor of Science (HBS), Mathematics and Computer Science <i>University of Toronto</i>	2021 – 2025 Toronto, ON
• Cumulative GPA: 4.00/4.00 (Average grade: 98.0%).	
• Governor General's Silver Medal (highest academic standing), John Black Aird Scholarship (top student).	

PUBLICATIONS

- **A.R.**, Jian Qian, Wenlong Mou. Adaptively Incorporating Directional Hints into Zeroth-Order Optimization. To appear in *Advances in Neural Information Processing Systems* (NeurIPS 2026). [arXiv:2609.08277](#).
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Convex Optimization with Delayed Feedback. To appear in *NeurIPS 2026*. [arXiv:2606.11711](#).
- **A.R.**, Idan Attias, Daniel M. Roy. A Reduction from Delayed to Immediate Feedback for Online Convex Optimization with Improved Guarantees. To appear in *NeurIPS 2026*. [arXiv:2602.02634](#).
- **A.R.**, Wenlong Mou. Reinforcement Learning with Action-Triggered Observations. In the *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. [arXiv:2510.02149](#).
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Learning with Delays: Scheduling Frameworks and Regret Trade-offs. In the *Proceedings of the 38th Conference on Learning Theory (COLT 2025)*, pp. 4973-5014. [arXiv:2503.19856](#).

PREPRINTS

- Idan Attias^{*}, Orin Levy^{*}, **A.R.**^{*}, Yishay Mansour, Uri Stemmer. Vanilla Policy Optimization Is Both Optimal and Differentially Private for Stochastic Contextual Bandits. [arXiv:2609.33888](#).

^{*}Equal contribution.

AWARDS

Connaught International Doctoral Scholarship (50,000 CAD)	2026 – 2031
Governor General's Silver Medal	2025
• Canada's most prestigious academic award, established in 1873; awarded to the undergraduate student with the highest academic standing upon graduation.	
John Black Aird Scholarship	2025
• Award recognizing the top graduating student at the University of Toronto.	
E. A. Robinson Medal in Sciences	2025
• Medal recognizing the top graduating science student at the University of Toronto.	
Vector Institute Research Grants (30,000 CAD)	2023 – 2024
University of Toronto International Scholar Award (180,000 CAD)	2021 – 2025
Undergraduate Scholarships & Research Awards at the University of Toronto	2021 – 2025
• MCS Research Award (7,500 CAD)	
• Joseph Alfred Whealy Scholarship (6,000 CAD total; four-time recipient)	
• Peter Fantham Memorial Scholarship in Mathematics (2,000 CAD)	
• Mariano A. Elia Scholarship (1,500 CAD total; two-time recipient)	
• Doris Geddes Scholarship in Mathematics (1,000 CAD)	

RESEARCH EXPERIENCE

Graduate Student Researcher	2023 – Present
<i>Vector Institute</i>	<i>Toronto, ON</i>
- Working on the theoretical foundations of modern AI: LLM post-training, scaling laws, optimization. - Completed Vector Institute Summer Research Internships (2023, 2024).	
Research Assistant	2022 – 2026
<i>University of Toronto, Department of Statistical Sciences</i>	<i>Toronto, ON</i>
- Studied the mathematical foundations of modern machine learning, with projects spanning feature learning in neural networks and high-dimensional statistical estimation. - Completed MCS Summer Research Internship (2022).	
Visiting Researcher	Spring 2023
<i>Sunnybrook Health Sciences Centre</i>	<i>Toronto, ON</i>
- Improved Swin UNETR for 3D segmentation in cancer and musculoskeletal research using dilated attention and a custom loss, outperforming the standard architecture in ablation studies. - Released the resulting implementation as an open-source codebase and presented the results at the Sunnybrook Orthopaedic Biomechanics Laboratory seminar.	

TECHNICAL SKILLS

Programming: Python, C/C++, Bash.
Machine Learning: PyTorch, JAX, Hugging Face Transformers.
Development Tools: Git, Linux, Slurm, Docker.

WORKSHOPS & TALKS

A Reduction from Delayed to Immediate Feedback for Online Convex Optimization. Poster.
<i>Learning Theory Summer School & Workshop</i> (LTSS 2026), Copenhagen, Denmark (Jun 2026).
Reinforcement Learning with Action-Triggered Observations. Seminar Talk.
<i>UofT Statistical Sciences Research Program</i> (UTSSRP 2026), Toronto, Canada (Jun 2026).
Capacity-Constrained Online Learning with Delays. Poster.
<i>Vector AI/ML “Remarkable 2026” Conference</i>, Toronto, Canada (Feb 2026). <i>Southern Ontario Learning Theory Workshop</i> (SOLT 2025), Waterloo, Canada (Nov 2025).

ACADEMIC SERVICE & TEACHING

Conference Reviewer	2026 – Present
- International Conference on Learning Representations (ICLR), 2027. - Conference on Neural Information Processing Systems (NeurIPS), 2026. - Conference on Learning Theory (COLT), 2026.	
Teaching Assistant	Spring 2025
<i>University of Toronto</i>	<i>Toronto, ON</i>
- CSC311, Introduction to Machine Learning. - CSC363, Computational Complexity and Computability.	